Project dossier — computational screening of sustainable Low-E PVD coatings

Everything found, why it was pursued, what it produced, and what it means.

Saint-Gobain Research India · Varun Mandy, Research Intern Framework: `pvdlowe` v0.1.0 · https://github.com/VarunMandy/pvdlowe 8,995 lines · 112 tests · 38 candidate architectures

How to read this document

Every section answers four questions in the same order: **what** was done, **why** it was worth doing, **what came out**, and **what it means**. Sections are ordered by how much they matter, not by when they happened.

The single most important caveat, stated once here and assumed throughout: no films were deposited. Every performance figure is a model output. The framework's own validation puts its median error at 30.6% against traceable literature, and its largest known error — an eightfold under-prediction of sputtered copper sheet resistance — sits in the material most leading candidates use.

PART I — THE ANSWER

1. What the project was asked

What. The brief `PVD_Usecase.docx` proposed a materials-by-design programme for low-emissivity coatings on float glass: a dielectric/metal/dielectric stack, AZO/Ag/AZO, with the question of how much silver could be removed while holding visible transmittance, sheet resistance and far-infrared reflectance.

Why. Silver dominates both the cost and the supply risk of Low-E glazing. At roughly 0.105 g/m² for a 10 nm layer it is the single largest materials cost in the stack, and its supply is concentrated and volatile.

Result. The question turned out to be underspecified in a way that changes the answer, and the framework's most useful output was identifying that.

Conclusion. See §2.

2. The answer depends on a question the brief did not ask

What. The brief specifies no climate, and the weighting derived from it does not constrain solar heat gain. That omission silently encodes a heating-dominated assumption.

Why it matters. Saint-Gobain Research *India*. If the target application is Indian architectural glazing, the unstated assumption is the wrong one.

Result. Adding the EN 410 / ISO 9050 solar-gain metrics and re-scoring produces two rankings that share **one candidate out of ten**.

	Heating-dominated	Cooling-dominated (India)
Lead candidate	$Si_3N_4/Ag_{10}Cu_{90}/Si_3N_4$ (E10e)	AZO/Cu/AZO (M6)
T_vis	0.877	0.738
Emissivity ε_h	0.0511	0.0463
Sheet resistance	3.13 Ω/sq	2.87 Ω/sq
Solar heat gain g	0.758	0.558
Silver	0.015 g/m² (**−86%**)	**zero**
Meets full spec	yes	**no** — T_vis 0.738 < 0.80

Benchmark for comparison — AZO/Ag/AZO at 10 nm: T_vis 0.876, ε_h 0.060, R_s 4.18 Ω/sq, 0.105 g/m² Ag, $0.85/m².

The mechanism is physical, not a scoring artifact. AZO is a *transparent conductive oxide*: its free carriers give a screened plasma wavelength at 1.25 µm, inside the solar band, so it rejects solar near-infrared. Si_3N_4 is a passive insulator, transparent throughout.

λ (nm)	1200	1600	2000	2400
Transmittance, AZO stack	0.240	0.115	0.068	0.046
Transmittance, Si_3N_4 stack	0.509	0.259	0.151	0.099

Same 11 nm copper layer in both. The conductive oxide is doing solar-control work the nitride cannot.

Conclusion. A results table is meaningless without its weighting file. **The climate must be specified before any recommendation is made.** For a heating climate the nitride stack at 5–10 at.% Ag beats the benchmark on transmittance, emissivity and sheet resistance simultaneously while using an eighth of the silver. For India, no single-metal architecture in the set meets the specification, and the framework is ranking near-misses.

3. The composition optimum is 5–15 at.% Ag, not 70%

What. The brief nominates $Ag_{70}Cu_{30}$ as the priority composition. Eight composition curves were run — Ag 0–100% in 5% steps, geometry re-optimised at every point, across two microstructure models, two dielectrics and two climates.

Why. If the optimum is elsewhere, the entire experimental programme is pointed at the wrong composition.

Result. Every one of the eight curves peaks between 0 and 10 at.% Ag.

Profile	Microstructure	Dielectric	Optimum Ag	Plateau
Heating	Segregated	Si_3N_4	**0.05**	0.00–0.20
Heating	Segregated	AZO	**0.10**	**0.05–0.40**
Cooling	Either	Either	0.00	0.00–0.15

And it is a sustainability result, not a performance one. Emissivity and sheet resistance keep *improving* up to 50 at.% Ag — 0.0386 and 2.59 Ω/sq, the best in the series. The score falls anyway because silver mass carries weight. The trade is about 0.005 in emissivity for an 87% cut in silver, and it is only the right trade because silver was decided to matter.

Plateau width matters more than peak position for a production process. The segregated/AZO curve is flat from 5 to 40 at.% Ag — a forgiving window.

Conclusion. $Ag_{70}Cu_{30}$ is outperformed in all eight combinations. Design at 5–15 at.% Ag. The brief's own justification for 70/30 traces to a 2022 paper that does support it — but on **polycarbonate**, for first-surface coatings, and §9 establishes that the underlayer determines metal grain structure, so that optimum does not transfer to a glass line unchecked.

4. Two metal layers break a ceiling one cannot

What. `MultiMetalCoating` generalises the architecture to n metal layers separated by n+1 dielectrics.

Why. The best light-to-solar-gain ratio across all 38 single-metal candidates is **1.37**, against roughly 2.0 for commercial solar-control glazing. Under the cooling profile no single-metal stack met the transmittance target at all. That is an architectural limit, and no compositional search could have found it.

Result.

Architecture	Metal	Dielectric	Geometry (nm)	T_vis	g	LSG	ε_h	R_s
Single	Cu	AZO	45/12/45	0.738	0.558	1.32	0.046	2.87
Double	**Ag**	**Si_3N_4**	**15/12/60/12/15**	**0.819**	0.464	**1.76**	0.025	1.67
Double	Cu	Si_3N_4	59/13/105/13/59	0.716	0.516	1.39	0.024	**1.33**

LSG 1.76 against a ceiling of 1.37 — 28% better.

Why one layer cannot. A single film trades transmittance against infrared reflectance along one curve; thickness moves along it but never off it. The dielectric *between* two metals adds an interference degree of freedom — the two reflections can cancel in the visible while adding in the infrared.

Note the optimum geometry: **thin outer dielectrics (15 nm), thick middle (60 nm)**. A symmetric guess misses it.

A third layer was tested and is not worth adding. LSG goes 1.33 → 1.76 → 1.81 for n = 1, 2, 3. The third buys 0.05 for a 25% silver increase, and for copper it is actively harmful — transmittance collapses to 0.437.

Conclusion. The architecture argument stops at two layers. Silver consumption doubles, since each layer must independently clear percolation near 10 nm, so double-Ag scores badly on the sustainability profiles despite being the best solar-control design found.

PART II — THE AUDIT

5. Four defects in the proposed weighting

What. The brief's §14 specifies criteria and weights. Encoding that table literally produced a scoring function that did not measure the project's stated objective. All four defects were found by the framework's own diagnostics (`pvdlowe check-weights`).

Why this matters more than it sounds. A scoring function is where every physical result gets converted into a ranking. A defect here corrupts everything downstream while leaving the physics untouched and every test passing.

Result.

(a) Silver consumption carried zero weight. §14's table has no line for it, although §17 and §20 both name minimising silver as an objective. Silver mass was computed, displayed, even reported as the limiting criterion — and contributed nothing to the score.

> The thickness optimiser exploited this immediately, returning a design using **29% more silver than the benchmark, at a perfect 100/100.**

(b) Emissivity and sheet resistance double-count. They correlate at **r = 0.996** across the candidate set — both are the free-carrier response of the same layer — so weights of 0.25 and 0.15 put 0.40 of the total on one physical quantity.

(c) Targets set at the specification minimum. Derringer–Suich desirability saturates at 1.0 once a target is reached. Setting T_vis's target to 0.80 — the brief's *floor of acceptability* — made the criterion flat exactly where candidates differ. Across 181 oxide pairs clearing 0.80, transmittance spanned 0.800–0.881 while the **score went down**, 74.99 to 73.30.

(d) The supply-risk band could not discriminate. At floor 8.0 against candidate values of 4.5–7.5, every silver-bearing composition scored 0.1–0.28.

A fifth issue: aggregation. The brief states its weighting should prevent a candidate "winning simply because it has excellent conductivity while being unacceptable optically". A weighted arithmetic sum does not achieve this — a candidate scoring zero on one criterion forfeits only that criterion's weight. A geometric mean does.

A fifth defect, found by re-auditing the fix for the second. Zeroing sheet resistance removed one double-count and left a triple-count untouched:

Pair	r
Silver mass ↔ metal cost	**1.000**
Silver mass ↔ supply risk	0.991

Silver dominates metal cost so completely that cost is the same quantity in different units, and supply risk in this candidate set *is* silver's supply risk. Together they carried **36% of the effective weight on one physical property**. Both are now weighted 0.0 and reported as derived figures; the effective concentration falls to 25%, and no criterion pair both carries weight and correlates.

And a residual that cannot be corrected, only disclosed. The silver weight of 0.15 is a judgement, not a measurement, and it selects the answer: the winner is a 0.065 g/m² design at weight 0, and silver-free at 0.30. Five of eight settings separate first from second by under 0.5 points, which means the ranking is resolving noise in a value judgement rather than distinguishing candidates.

`weight_sweep()` reports the transitions instead of a single ranking. **The honest recommendation is: E10e at weight 0.15, E5e at 0.20–0.25, and the silver-free N6 at 0.30 and above — and choosing among them is a decision about how much silver consumption matters to Saint-Gobain, not one the framework can make.**

A sixth, found by asking what the framework does not predict. Three criteria carried 0.30 of the 0.90 nominal weight while being `None` for every candidate, so a third of the weight renormalised away silently — a file stating emissivity at 0.25 was applying 0.42. One of them, `structural_stability`, had no criterion definition at all and could never have scored. All three weighted 0.0; the ranking is unchanged, which is the point.

`structural_stability` is now partly populatable from the Materials Project screening and the surrogate mixing energies, and is deliberately left unpopulated: doing so would change every ranking on values carrying 20–40% systematic error, at a weight nobody derived. Populating a criterion and choosing its weight are one decision, and it belongs to whoever continues.

Conclusion. Four defects corrected, two more found and corrected on re-audit, and one residual disclosed rather than papered over. None applied silently. **General rule extracted: a desirability target should be an aspiration, not a specification minimum — and a weight that changes the winner should be reported as a sweep, not a value.**

6. The evidence base: eight of fourteen claims changed

What. Every citation in the brief carried `utm_source=chatgpt.com`, meaning the sources were surfaced by a language model rather than read from a publisher. All fourteen claims were checked.

Why. Not because LLM-surfaced means wrong — several are reproduced well by the model, which is weak evidence they are right — but because none had been verified, and two carry load-bearing conclusions.

Result.

Status	Count
Verified — full text read	4
Partial — source located, abstract confirmed	6
Disputed — figures appear in no locatable source	**2**
Supported — corroborated independently	3
Not located after searching	1

Two citations cannot be supported. The figures 85.4% T_vis / 3.21 Ω/sq / 97% FIR, and 78.7% T_vis / 2.7 Ω/sq, appear in neither located AZO/Ag/AZO study. **Do not cite them.**

> **An uncomfortable corollary.** These two are the model's three *closest* agreements — 0.3%, 1.0%, 1.2% relative error. Excluding them, the median validation error rises from **14.7% to 30.6%**. The framework's apparent accuracy rested partly on figures that cannot be traced. `validate_model()` now excludes them by default, and a test fails if that exclusion ever starts *lowering* the reported error.

Two transcription errors found. The brief states Ag = 1.59 μΩ·cm for a 10 nm film; the source patent (US 10,822,692) says **1.29**. Separately, an earlier "correction" of mine — that the brief's 82.4% AZO transmittance should be 81.4% — **was itself wrong**: the paper reports both, 82.4% in Results and 81.4% in the abstract, and the brief quoted Results.

Three factual corrections. Deposition methods differ across the benchmark set — RF for the AZO/Ag/AZO trilayers where §3 implies DC, medium-frequency for the AZO single layer — so these sources cannot be pooled.

Conclusion. Checking citations changed eight of fourteen. That is a finding in itself, and the mixed table is more informative than a clean one would be.

7. A measurement that contradicts a model-independent limit

What. *Applied Surface Science* **578** (2022) reports AZO/Cu/AZO with T_vis 87.7%, R_s 9.96 Ω/sq and ε 0.055. **The brief's §16 conclusion — that a silver-free stack may already meet a Low-E target — rests entirely on it.**

Why. For a conducting sheet much thinner than the wavelength, far-infrared reflectance is fixed by sheet resistance alone:

```
r = (n₁ − n₂ − Z₀/R_s) / (n₁ + n₂ + Z₀/R_s),   ε = 1 − r²
```

with Z_0 = 376.73 Ω. **No fitted parameter enters.**

Result. At 9.96 Ω/sq the floor is **ε ≥ 0.096**. The reported 0.055 would require about 5.5 Ω/sq. Five explanations were tested and eliminated:

Explanation	Outcome
Different samples	Ruled out — abstract attributes all three to one sample
Transcription error	Ruled out — confirmed against the abstract
Far-IR glass index assumption	Ruled out — limit is 0.091–0.097 for n = 1.5–4.0
Band-limited emissometer	**Ruled out — reads 5-8% *higher*, moving it the wrong way**
The framework's own convention	**Ruled out — an industrial formula agrees**

That last one matters most. Cueva & Carretero compute emissivity from sheet resistance as ε = 0.0106·R□, citing Gläser, *Large Area Glass Coating* (2000) — in industrial use for two decades. It agrees with the framework's impedance limit within 10% and gives **0.106** at 9.96 Ω/sq against the reported 0.055.

The same group reports the same ~0.6 ratio in two further papers, so it is systematic rather than an isolated error.

Conclusion. The claim has hardened from "unverified" to **"inconsistent with a model-independent electrodynamic limit, with every benign explanation tested and eliminated"**. Untested: hemispherical versus normal basis, which would also *raise* the reading. **§16's conclusion should not be used until this is settled** — one line of a paywalled full text.

PART III — WHERE THE MODEL WAS WRONG

8. The dielectric finding was reversed by measurement

What. The framework originally concluded that Si_3N_4 outperforms AZO on transmittance by up to nine points, on index-matching grounds.

Why it was tested. Cueva & Carretero (*Coatings* **13**, 1709, 2023, open access) deposited five dielectrics under identical conditions with 10 nm Ag.

Result — the opposite.

Dielectric	Measured ε	n(550 nm)
SnO_2	0.083	2.00
ZnO	0.064	2.019
AZO	**0.058**	**1.85**
SiAlNx	0.067	2.09

AZO wins on emissivity *and* transmittance despite having the lowest refractive index of the five. The authors give the reason: silver growth is more efficient on AZO.

What the model was missing. It treated the dielectric purely as an interference layer — index and thickness, nothing else. The optical inputs were sound; their measured indices match the framework's closely. What was absent is that the underlayer determines the *quality of the metal grown on it*.

The correction. `TCOPreset.metal_growth_factor` applies a resistivity penalty per underlayer, calibrated to that series and normalised to AZO = 1.00: ZnO 1.10, SiAlNx 1.16, ITO 1.20, TiO_2 1.25, SnO_2 1.43. The model now reproduces the measured ordering and magnitudes to 4–8%.

Conclusion, and the productive outcome. The brief's framing did exclude nitrides — it began from the periodic table and the Materials Project, both of which surface oxides — and that observation stands. But the best answer is a **hybrid**, which is what industry already does: AZO beneath for nucleation, nitride above for durability through tempering. Two candidates were added:

ID	Stack	T_vis	ε_h	Heating
H1	AZO/Ag/Si_3N_4	**0.918**	0.057	67.4
M0	AZO/Ag/AZO (benchmark)	0.876	0.060	65.0
H2	AZO/Cu/Si_3N_4 (silver-free)	0.788	**0.042**	79.9

H2 now ranks **second on the cooling profile** and is the only architecture appearing in both top-ten lists.

9. The mechanism, and the framework's principal structural weakness

What. Two ML surrogate calculations were run to find the mechanism behind `metal_growth_factor`. Both failed. A literature search then found it in under an hour.

Why it was worth chasing. An empirical multiplier with no mechanism is a fitted fudge factor. With one, it becomes a physical parameter that can be measured and predicted.

Result — the failures, both diagnosed:

Attempt	Failed on	Diagnostic that caught it
Bulk interface adhesion	5–21% lattice mismatch — measured elastic strain	unphysical outputs: 9.6 and −0.16 J/m^2
Adatom wetting	ZnO(0001) termination changed the answer by 2.1 eV against 0.46 eV between materials	explicit termination sweep

Result — the mechanism. US 7,632,572 B2 / US 8,512,883 B2 (AFG Industries → AGC Flat Glass North America → Cardinal CG; **full text read**) deposited 16 nm Ag on amorphous TiOx and on a 5 nm ZnO seed over the same TiOx, and examined both by TEM:

- **25 nm grains on ZnO against 15 nm on a-TiOx**
- {111}-oriented grains **two to three times larger** on ZnO
- the film on bare amorphous titania **clearly discontinuous** where the ZnO-seeded film was continuous across the whole specimen

And the inventors name the mechanism: zinc oxide grows {0001}, which orients the silver to grow {111}, and the **epitaxial**

lattice match between Ag{111} and ZnO{0001} lowers sheet resistance and improves adhesion.

An unexpected validation. The patent also reports four-point sheet resistance: **5.68 Ω/□ with the ZnO seed against 7.56 without**, a ratio of **1.331**. `metal_growth_factor` — calibrated independently from Carretero's *emissivity* series, a different group, decade and measured quantity — gives **1.250**. **Agreement to 6%.** A parameter fitted to one dataset reproducing another it never saw.

The conclusion, and it is the most consequential in the project:

> **The framework models each metal layer as though the layer beneath it did not shape its microstructure.** That single omission accounts for *both* `metal_growth_factor` *and* the eightfold under-prediction of sputtered copper sheet resistance. They are one effect appearing twice, not two caveats.

The default `grain_size_ratio` of 3.0 corresponds to 30 nm grains in a 10 nm film — within 20% of the 25 nm the patent measured on ZnO. The assumption was correct, but only for the underlayer it happened to be tuned against.

10. Corrections to the project's own conclusions

What. Four conclusions stated during this work were later overturned by the work itself. They are recorded rather than quietly fixed.

Why. A methods section that shows its own corrections is more credible than one that does not, and the common cause is transferable.

Claim	Overturned by
An optimum exists at $Ag_{60}Cu_{40}$	A three-point artifact. At 5% resolution the curve is monotonic
The Ti barrier variant beats the ternary	An artifact of silver carrying zero weight. Corrected, it ranks last
The band-emissometer explains the ApSS emissivity	Implemented and tested: it reads 5–8% *higher*, worsening the discrepancy
Adatom wetting reproduces the measured ordering	A termination sweep showed it was resolving slab index, not material

Conclusion. Three of four are the same error — **reading structure into too few points**. Three compositions in one case, one geometry per composition in another, one untested hypothesis in a third. The framework's value came largely from re-running things at finer resolution and watching earlier conclusions dissolve.

PART IV — WHAT WAS COMPUTED, AND WHAT IT COST

11. Thermodynamics: Ag–Cu has no stable ordered compound

What. All fifteen chemical systems from the brief's §9 were retrieved from the Materials Project and screened within 50 meV/atom of the convex hull.

Result. Two intermediate Ag–Cu phases exist and **neither is stable**: Cu_3Ag at 0.0904 eV/atom above the hull, $CuAg_3$ at 0.0857. Ag–Cu is the only nominated binary with no stable intermediate phase.

The magnitude matters: 86–90 meV/atom is well above k_BT at deposition (26 meV at 300 K, 48 meV at 550 K), so the driving force to separate is substantial.

A second finding, bearing on the Ti hypothesis. Cu–Ti has 8 near-hull phases and Al–Cu has 10. A dilute Ti or Al addition to a Cu-bearing film therefore has stable intermetallics available to precipitate rather than remaining in solution as a stabiliser — a second, independent mechanism for the ternary underperforming, alongside the Nordheim conductivity penalty.

Conclusion. The brief's §5 suspicion of phase separation, inferred from a 13% lattice mismatch, is now supported by first-principles data.

12. Mixing energies: the dilute corner is thermodynamically favoured

What. MACE-MP-0 and CHGNet on 32-site fcc supercells, three random decorations per composition, gated against the two known hull distances.

Why. The convex hull is an equilibrium statement about *ordered* phases. A sputtered film is neither ordered nor at equilibrium, so the disordered solid solution needed separate treatment.

Result. Positive across the whole range, fitting a regular-solution form to ±6 meV/atom: **ΔE_mix = 0.287·x·(1−x)**.

A self-check that matters more than the gate. At Ag 25 at.% the surrogate puts the *disordered* solution 13 meV/atom **below** the ordered Cu_3Ag compound. No ordering tendency at all — precisely what an empty hull implies, reached independently.

The finding that bears on the design:

Ag at.%	ΔE_mix	×kT (550 K)
70 — the brief's priority	0.0602	1.27
15	0.0366	0.77
10	0.0258	0.54
5	0.0136	0.29

In the dilute-silver corner the driving force to separate falls below thermal energy at deposition. §3 found the two microstructure hypotheses converge there and treated it as robustness; the thermodynamics now supply the reason.

The dilute-silver optimum is therefore supported on three independent grounds: lowest silver consumption, highest score under the corrected weighting, and the weakest tendency to phase-separate.

Cross-checked, with a caveat. Both models under-predict — MACE by 0.026, CHGNet by 0.047 eV/atom — so the bias is inherited from shared Materials Project training data rather than architecture-specific, meaning ΔE_mix should be corrected *upward*. **They are not independent**: both are trained on the same relaxation trajectories, so agreement shows the bias is consistent, not absent.

And a compounded correction, from verifying one of the brief's own citations. GGA underestimates Cu–transition-metal formation energies by nearly 40%, because it places Cu-3d bands too shallow (npj Comput. Mater. 2024). The MP hull is GGA/GGA+U and both surrogates train on it, so two errors act in series. The conclusion **holds at 5–10 at.% Ag and becomes marginal at 15%**.

13. What the surrogate could not do

Cannot, at all: optical constants, dielectric functions, band gaps, emissivity. These models predict energies and forces from atomic positions and carry no electronic structure. **Every optical number in this project must come from the transfer-matrix model or from measurement.**

Unreliable: amorphous or defective phases, metal/oxide interfaces, and any energy difference below about 30 meV/atom — which is comparable with the hull distances being measured.

PART V — WHAT REMAINS

14. The critical unknown

What. The framework validates well against silver in the infrared — within 0.4% on the benchmark it reproduces best — and **fails on copper**, under- predicting sputtered sheet resistance by roughly eightfold.

Why it matters. Six of the leading candidates in both climate profiles are copper-based.

Result of the literature check. Four independent published fits of the scattering parameters give specularity 0.48–0.80 and grain-boundary reflection 0.16–0.43. The framework's assumed 0.50 and 0.25 sit inside that range, and **an exhaustive scan over the entire admissible range reaches only 4.42 Ω/sq against the 16.6 to be explained** — short by a factor of 3.8.

So the classical size effect is eliminated, and grain structure is the leading hypothesis: grains at roughly half the film thickness give 14.1 Ω/sq.

Conclusion. If that reading is right, **copper's obstacle is film quality, not physics** — an engineering problem with known levers (base pressure, seed layers, substrate temperature) rather than a fundamental limit. That is far more tractable than replacing silver's electronic structure.

15. The experiment that resolves it

One XRD scan, one film, an afternoon — and it answers three questions.

`pvdlowe/characterise/xrd.py` states in advance what the scan should show:

Measurement	Question answered
Peak count and position	**Microstructure.** Segregated gives two fcc sets at 38.12° and 43.32°; a solid solution gives one at 39.54° — 1.42° from silver against a 0.59° width
Scherrer peak width	**Grain size**, and therefore `grain_size_ratio` — the framework's principal structural weakness
(111)/(200) intensity ratio vs the powder value of 2.2	**Templating**, the mechanism of §9

Scan to 2θ = 100°, not 60. Separation grows faster than width with angle: the (222) reflection separates at 4.8× its width against (111)'s 2.4×.

A second, independent discriminator. Sheet resistance separates the microstructure hypotheses by a number — 6.4 against 2.6 Ω/sq. Diffraction separates them by *peak count*. Two unrelated observables agreeing is far stronger than either alone, and both come from one film.

Pre-registered decision rules, written before any data exists, in `experiments/PROTOCOL_cu_series.md`. Four outcomes, each with what it means and what to do — **including the one where the literature is right, six candidates fall, and the project reverts to silver reduction.**

16. The open question only SGRI can answer

What. The report benchmarks against a 10 nm AZO/Ag/AZO at ε_h 0.060, taken from the brief — whose citations for that stack are the two **disputed** entries.

Why it matters. Saint-Gobain's own patents (US 7,745,009) claim silver at **12.5–16 nm targeting ε ≤ 0.038**. The model says that needs 14–16 nm.

Ag nm	ε_h	Ag g/m²
10.0 (current benchmark)	0.060	0.105
14.0	0.039	0.147
16.0	0.033	0.168

Conclusion. If production runs at 12.5–16 nm, the reported silver reductions are measured against a stack that would not meet SGRI's own emissivity specification. **The baseline needs confirming before the percentages are quoted.** This is the single highest-value question outstanding and only SGRI can answer it.

PART VI — WHAT WAS BUILT

17. The framework

8,995 lines, 112 tests, ten subpackages, nine CLI commands.

The design decision that matters. Film resistivity feeds the Drude damping, so a metal layer too thin to conduct is automatically one that reflects poorly. Without that coupling, optimising for transmittance drives silver thickness to zero and the model reports an excellent coating.

Validated against closed-form answers, not by inspection: exact Fresnel for a bare interface, exact null for a quarter-wave layer, energy conservation to 10^{-10}, s and p agreeing at normal incidence.

Provenance as a type, not a footnote. Every quantity carries an evidence grade, and a guard refuses HYPOTHESIS-grade values in headline tables. This is what kept the Ti-ternary predictions from being quoted as results.

Functions that refuse rather than guess. The sputter model raises when uncalibrated; the MP client raises when offline with a cold cache; the surrogate refuses to fall back to an empirical potential; the adhesion calculation refuses above 4% lattice mismatch. Each was tempting to make helpful; each would have produced fabricated data.

18. Known limitations, stated plainly

No experimental validation. Every performance figure is a model output.

Fitted parameters: specularity and grain-boundary reflection are fitted, not measured. Si_3N_4 and TiO_2 optical constants are ESTIMATE grade.

Illuminant approximation, now quantified: varying colour temperature over 5000–7500 K moves **T_vis by 0.1%** and **T_sol by 8.3%**. T_vis is reportable as computed; T_sol, g and LSG carry a 5–10% systematic until tabulated AM1.5G spectra are supplied.

Scores are not portable. Absolute scores moved by ~25 points as the §5 defects were fixed. A score is not a property of a material.

Not modelled at all: roughness, interdiffusion, damp-heat and abrasion durability, adhesion, agglomeration kinetics, and any property of a metastable sputtered alloy that a 0 K ordered-crystal database cannot supply.

CONCLUSION

What this project established

A better composition. 5–15 at.% Ag rather than 70%, supported on three independent grounds — silver consumption, corrected score, and thermodynamic stability against segregation.

A better material class. Silicon nitride was outside the brief's search framing; the hybrid AZO/metal/Si_3N_4 arrangement beats both pure stacks and is what industry already does.

A better architecture. Two metal layers reach LSG 1.76 against a single-metal ceiling of 1.37 — an architectural limit no compositional search could have found.

A specified climate dependency. The two profiles share one candidate out of ten. The brief's silence on climate was a substantive omission, not a detail.

What this project corrected

Four defects in the proposed weighting, one of which let the optimiser return **29% more silver than the incumbent at a perfect score**. Two citations that cannot be traced. Two transcription errors. One measurement inconsistent with a model-independent physical limit. And four of its own conclusions, overturned by finer resolution and recorded rather than hidden.

What this project cannot claim

Nothing has been deposited. The candidate rankings are hypotheses ordered by physics, suitable for directing experimental effort. Their principal weakness — that the framework models each metal layer as though the layer beneath it did not shape its microstructure — is known, quantified, and addressed by a specified experiment.

The recommendation

One XRD scan, one film, an afternoon. It measures the grain size that §9 identifies as the common cause of two separate limitations, tests the templating mechanism directly, and discriminates the microstructure hypotheses by a second independent route.

And **one question to SGRI**: which stack is the right baseline.

> A model that identifies the measurement capable of falsifying it is more useful to an experimental programme than one that does not. That is what this framework is for, and it is what it delivered.